

PeriNet: Periodic Deep Learning Framework For Modality-Agnostic Privacy Preserving

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Abstract

The proliferation of Web-based AI services has necessitated the outsourcing of massive user data to cloud platforms, raising critical privacy concerns regarding data transmission over public networks. Existing privacy defenses often struggle to balance the strict privacy requirements of open web environments with the high utility and low latency demanded by real-time applications. To address these challenges, we propose *PeriNet*, a modality-agnostic Privacy-Enhancing Technology (PET) designed to secure client-to-cloud data interactions. On the web client side, *PeriNet* employs *Gaussian Periodic Obfuscation* to inject structured periodic noise, statistically decoupling sensitive user payloads from their raw semantics before transmission. On the cloud server side, we introduce a *periodic deep neural network* that embeds periodicity throughout the architecture. Specifically, nonlinear operators are replaced by periodic activation functions, and linear layers are constrained by periodic regularization to process obfuscated web data without decryption. Furthermore, a *Periodic-aware Contrastive Learning* scheme is adopted to accelerate convergence. Extensive experiments on multimedia benchmarks (covering CNNs and Transformers for vision and text) demonstrate that *PeriNet* provides robust protection against web-based reconstruction and inversion attacks by exponentially enlarging the adversarial search space. It achieves up to 99.34% accuracy in vision tasks and 94.1% in language tasks, outperforming state-of-the-art privacy-preserving baselines by up to 8%, making it a practical solution for secure Model-as-a-Service (MaaS) deployment.

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CCS Concepts

• Security and privacy; • Computing methodologies → Artificial intelligence; Machine learning;

Keywords

Periodic Privacy Preservation; Modality-Agnostic; Gaussian Periodic Noise; Contrastive Learning.

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1 Introduction

In the era of the Intelligent Web, the seamless integration of deep learning into web-based services has revolutionized applications ranging from visual search engines to large language models (LLMs), necessitating the processing of massive-scale datasets [1]. Due to the computational constraints of edge devices, data owners increasingly rely on Web-based Model-as-a-Service (MaaS) paradigms and public cloud infrastructures to execute intensive training and inference tasks [22, 27, 43]. However, this outsourcing of computation via web protocols introduces critical security and privacy vulnerabilities, as sensitive user data transmitted to semi-trusted cloud providers becomes exposed to potential server-side breaches and adversarial interception [17, 23, 35]. These privacy risks are exacerbated in high-stakes web applications, such as telemedicine platforms where image reconstruction attacks can leak patient records [45], or online financial services where model inversion can extract confidential entities from uploaded documents [10, 37].

To mitigate these threats, Privacy-Enhancing Technologies (PETs) for web-based machine learning have evolved from rudimentary anonymization to sophisticated training-time defenses [7, 15, 44].

As the Web ecosystem moves towards open data sharing and collaborative intelligence, safeguarding multimedia content during outsourced online computation has become a pivotal challenge in the security and privacy of AI applications. In the context of visual web content, representative approaches include generative surrogates [11], structural transformations (e.g., Disguised-Nets [32], VisualMixer [21]), and differential privacy mechanisms [8, 38]. While these methods obscure identifiable details, they often incur high latency or semantic distortion, rendering them less suitable for real-time web services. Similarly, for textual web data, defenses focus on mitigating semantic leakage via differentially private fine-tuning [2, 41], token replacement (e.g., 1-Diffractor) [28], or embedding perturbation [7]. However, these modality-specific techniques frequently degrade the utility required for high-precision web tasks. Furthermore, existing solutions lack the cross-modality generalization needed for the diverse, unstructured data types prevalent on the modern Web.

Inspiration. Existing noise-injection methods often result in significant performance degradation in downstream tasks, since DNNs, as highly nonlinear and nonconvex systems, amplify even minute input perturbations through layer-wise propagation, leading to unstable outputs and distorted feature representations. Such instability exposes a fundamental limitation of noise-based obfuscation and motivates the pursuit of transformations with intrinsic regularity and bounded responses [19, 30]. Periodic functions [9], characterized by their inherent *translation invariance under integer-period shifts* ($f(x) = f(x + T)$), produce identical outputs for periodically shifted inputs. This property enables structured perturbations to be absorbed within the same representational cycle, constraining feature fluctuations and mitigating noise sensitivity. By embedding periodicity into both data obfuscation and network representation, feature responses can be regularized within controllable ranges, thereby enhancing robustness against inversion and reconstruction attacks while preserving expressive capacity. This observation inspires our design of a periodic deep learning framework that leverages periodic transformations to achieve modality-agnostic privacy preservation.

Challenges. Leveraging periodic functions introduces further challenges. First, it remains non-trivial to inject periodicity into DNN architectures (e.g. CNNs and Transformers) while maximally preserving their original expressive power and representation capacity. Second, the oscillatory nature of periodic functions can hinder gradient-based optimization, causing unstable convergence or vanishing gradients during backpropagation. Addressing these challenges is essential to effectively harness the benefits of periodic transformations for privacy-preserving learning.

Solutions. To fundamentally balance privacy preservation and model utility, we propose *PeriNet*. Instead of relying on random noise or cryptographic defenses, *PeriNet* leverages the invariance property of periodic functions to make sensitive information cyclically indistinguishable while retaining task-relevant semantics. Specifically, a Gaussian-periodic obfuscation mechanism introduces structured periodic perturbations that fold privacy-sensitive features into a cyclic manifold, concealing local identity cues without degrading the global information required for downstream tasks. The core network replaces conventional nonlinear activations with the periodic activation function SinLU and imposes periodic

weight constraints, ensuring that inputs differing by a periodic shift are mapped to equivalent representations—thereby neutralizing instance-specific leakage at the representational level. Furthermore, a periodicity-aware contrastive learning paradigm aligns latent representations of different periodic variants, reinforcing semantic consistency under structured perturbations and enhancing generalization in privacy-preserving scenarios. In summary, the main contributions of this paper are as follows.

- We analyze the degradation mechanism of existing noise-injection methods, showing that the inherent sensitivity of deep nonlinear networks amplifies small perturbations into unstable and distorted feature representations.
- We propose *PeriNet*, which integrates a theoretically grounded Gaussian-periodic obfuscation mechanism with a periodic DNNs architecture, achieving privacy preservation without sacrificing task performance.
- We demonstrate its modality-agnostic versatility, showing applicability across both visual and textual data, and seamless integration into mainstream architectures such as CNNs and Transformers.
- We validate *PeriNet* on diverse datasets and modalities, achieving strong privacy protection with modest accuracy loss and low overhead, while resisting representative adversarial attacks, reaching up to 99.34% accuracy in vision tasks and 94.1% in language tasks, outperforming the strongest existing privacy-preserving baselines by up to 8%.

The source code will be publicly available at <https://github.com/Aurelius-04/Perinet>.

2 Related Work

To contextualize our approach, we summarize representative privacy-preserving obfuscation methods across modalities in Table 1. Existing techniques are broadly divided into vision and text modality obfuscation methods, which vary in modality assumptions, privacy mechanisms, and computational trade-offs.

Vision Modality Obfuscation. In the vision domain, privacy-preserving research has explored a variety of obfuscation strategies to mitigate the risk of visual data leakage before outsourced computation. Generative approaches, such as PPGAN [25], synthesize surrogate images that preserve the semantic content of the originals while obscuring identity-related attributes. However, these methods often suffer from semantic bias and distributional shift. Structural transformation techniques take a complementary route. For instance, Disguised-Nets [32] employ block permutation, multi-dimensional transforms, and additive noise to enhance protection, but incur substantial computational overhead at scale. VisualMixer (VIM) [21] reduces complexity by shuffling pixels without noise addition, thereby maintaining model accuracy but lacking formal privacy guarantees. More recently, CODER [38] adopts a metric differential privacy framework by injecting noise in a hybrid space defined by pixel-level and histogram-based distance metrics, achieving provable privacy and high-fidelity image protection. Nevertheless, its noise sampling process remains computationally expensive, and the choice of key parameters is nontrivial, limiting its efficiency and practicality.

Table 1: Comparison of representative obfuscation strategies with PeriNet.

Method	Attributes		
	Modality Generalization	Privacy Guarantee	Efficiency
PPGAN [25]	○	●	○
Disguised-Nets [32]	○	●	●
VIM [21]	○	●	●
CODER [38]	○	●	○
Ew-tune [2]	●	●	○
1-Diffractor [28]	●	●	●
EP [7]	●	●	●
PeriNet	●	●	●

“●” denotes *Fully covered*, “●

Text Modality Obfuscation. In the text domain, research targets linguistic and semantic leakage in model training and inference. Gradient-based methods like Ew-tune [2] add calibrated noise to clipped gradients, achieving formal privacy at the cost of disrupted optimization and reduced accuracy. Replacement strategies such as 1-Diffractor [28] substitute sensitive tokens under local differential privacy, improving efficiency but risking semantic distortion. Embedding perturbation (EP) [7] injects noise directly into embeddings to resist inversion attacks, though excessive perturbation degrades downstream performance. Overall, these approaches alleviate privacy leakage to varying degrees but struggle to balance protection, utility, and robustness.

As summarized in Table 1, most existing obfuscation strategies are modality-specific—vision models rely on spatial transforms, while text models depend on linguistic regularities. Cross-modal approaches like Ew-tune provide formal guarantees but at the cost of significant accuracy loss. Conversely, heuristic methods such as pixel shuffling or token replacement improve efficiency but lack theoretical rigor. These limitations motivate the need for a *unified, modality-agnostic* framework that offers provable privacy, computational efficiency, and utility preservation. Our work introduces periodic transformations as a mathematically grounded mechanism to jointly enhance stability, privacy, and efficiency.

3 Threat Model

We consider a widely adopted web-based MaaS architecture, where data owners (acting as web clients) locally obfuscate sensitive inputs using *Gaussian Periodic Obfuscation* before transmitting them to a cloud-based AI service for training or inference. In this client-server web interaction, the client (e.g., a local browser or edge device) is fully trusted and never exposes raw private data to the network. Conversely, the cloud service provider acts as an honest-but-curious adversary [42]. This implies the server faithfully executes the web API protocols and computation tasks but may covertly attempt to harvest, reconstruct, or correlate private user information from the received obfuscated payloads or model outputs. The adversary possesses full access to the server-side environment, including the transmitted data streams, model parameters, and gradient signals. They may employ advanced reconstruction-based attacks—such

as feature inversion, generative reconstruction, or inverse mapping—to compromise user privacy. *PeriNet* is designed to secure this web-based data exchange, preventing meaningful recovery of raw sensitive attributes while preserving the utility and efficiency required for real-time web applications.

4 PeriNet: Methodology

4.1 Overview

The *PeriNet* framework achieves privacy preservation by embedding periodic invariance into both data transformation and model representation, enabling protection rather than post-hoc defense. Unlike conventional noise-based or encryption-based methods that rely on external perturbation, *PeriNet*, as shown in Figure 1, restructures the representation space so that sensitive information is cyclically invariant—rendering private patterns unrecoverable while maintaining task-relevant semantics. The framework integrates three mechanisms: a Gaussian-periodic perturbation at the client side introduces structured periodic noise that folds local privacy features into a cyclic manifold; the server-side model employs the periodic activation and periodic weight constraints to ensure equivalent responses for inputs differing by a periodic translation, suppressing instance-specific identity signals; and a periodic contrastive regularization enforces consistency across representations under distinct periodic perturbations, aligning latent spaces along invariant dimensions while preserving inter-sample separability. Through this hierarchical periodic alignment, *PeriNet* transforms privacy preservation into a representational property of the model itself, effectively decoupling privacy-sensitive variations from semantic information and ensuring that adversarial reconstruction yields semantically neutral outputs without compromising utility. The complete training pipeline of *PeriNet* is provided in Algorithm 1.

4.2 Gaussian Periodic Obfuscation

This section presents an efficient obfuscation mechanism that aligns inputs with Gaussian noise through element-wise periodic shifts, effectively concealing sensitive information while preserving the original semantic structure with minimal distortion. The core idea is to shift each input element by integer multiples of the minimal positive period defined by the periodic neural architecture, ensuring that the obfuscated data distribution approximates Gaussian statistics while maintaining interpretable periodic representations.

Specifically, let $\mathbf{X}$ denote the input tensor (the original image or the embedded form of other input modality) and $\mathbf{S} \sim \mathcal{N}(\mu, \sigma^2)$ denote a sampled Gaussian noise reference tensor of the same shape. Here, σ is the standard deviation, and μ is the distribution mean. The obfuscation formula is as follows:

$$\mathbf{X}' = \mathbf{X} + \mathbf{N} \cdot \mathbf{T}, \quad (1)$$

where $\mathbf{N} \in \mathbb{Z}^{\text{shape}(\mathbf{X})}$ is an optimal integer matrix whose elements n are computed as follows:

$$n = \arg \min_{m \in \mathbb{Z}} |x + m \cdot T - s|, \quad (2)$$

where x and s represent the corresponding elements of $\mathbf{X}$ and $\mathbf{S}$, respectively. And m is an arbitrary integer. To illustrate how the unified mechanism can be instantiated across different data

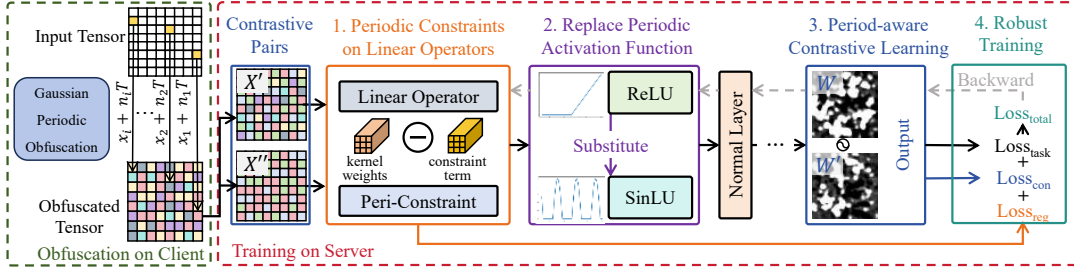


Figure 1: Overview of the proposed PeriNet framework. The system operates in a client–server architecture, where the client performs Gaussian Periodic Obfuscation to inject structured periodic noise into input data, producing obfuscated tensors that statistically decouple private content. On the server side, training incorporates three key modules: (1) periodic constraints on linear operators (Peri-Constraint) to align weights with periodic lattices, (2) replacement of traditional ReLU with the proposed SinLU activation to stabilize periodic responses, and (3) periodic-aware contrastive learning to ensure representation consistency across periodic variants. These modules jointly optimize the total loss composed of task loss, contrastive loss, and regularization loss, achieving privacy preservation with robust model performance across modalities.

modalities, we further detail its application across the image and text modalities as follows.

Image Modality Obfuscation. As shown in Appendix D Figure 4, let the input be an image represented by a pixel matrix $I \in \mathbb{R}^{w \times h}$, where w and h denote the width and height. The obfuscation transforms I into $I' \in \mathbb{R}^{w \times h}$ such that its pixel values align with samples from a Gaussian distribution. Specifically, a reference matrix S is sampled with the same shape as I , and the obfuscation is given by

$$N_{\text{img}} = \arg \min_{M \in \mathbb{Z}^{w \times h}} \|I + M \cdot T - S\|_F, \quad (3)$$

$$I' = I + N_{\text{img}} \cdot T, \quad (4)$$

where T is the minimum positive period of the constructed periodic DNNs and $\|\cdot\|_F$ denotes the Frobenius norm. Each obfuscated pixel approximates a Gaussian sample, yielding distributional alignment. This per-pixel independence generates globally complex patterns that resist inversion, as shown in Appendix D Figure 5, thereby enhancing adversarial robustness and reducing the risk of inferring original characteristics.

Text Modality Obfuscation. Analogous to the image modality, the input text is represented by an embedding matrix $E \in \mathbb{R}^{L \times d}$, where L is the sequence length and d the embedding dimension. A reference matrix S is sampled, and the optimal integer matrix $N_{\text{text}} \in \mathbb{Z}^{L \times d}$ is computed as

$$N_{\text{text}} = \arg \min_{M \in \mathbb{Z}^{L \times d}} \|E + M \cdot T - S\|_F, \quad (5)$$

leading to the obfuscated embedding matrix

$$E' = E + N_{\text{text}} \cdot T. \quad (6)$$

This transformation produces embedding values that differ substantially from the original while statistically resembling random noise, as shown in Figure 2, making it infeasible to reconstruct the raw embeddings or extract semantic content.

As validated in Section 5.4, the obfuscation induces a combinatorially large brute-force search space and empirically thwarts both statistical reconstruction and GAN-based inversion attempts. Computationally, the method remains lightweight and parallelizable: (i) each element update $x' = x + n \cdot T$ requires only integer arithmetic;

(ii) the optimal n is obtained in closed form without iterative solvers; (iii) independence across elements enables GPU-level parallelism. For N total elements per sample, the effective cost is $O(N)$ under parallel execution, supporting real-time, high-resolution processing.

$$\begin{bmatrix} 1.24 & 0.89 & 1.56 \\ 0.91 & 1.12 & 0.67 \\ 1.33 & 0.98 & 1.21 \end{bmatrix} = \begin{bmatrix} 0.24 & -0.11 & 0.56 \\ -0.09 & 0.12 & -0.33 \\ 0.33 & -0.02 & 0.21 \end{bmatrix} \begin{matrix} 1 \\ \text{love} \\ \text{WWW26'} \end{matrix} + T \begin{bmatrix} 1 & -2 & 0 \\ -1 & 1 & 2 \\ 2 & 0 & -1 \end{bmatrix}$$

E' (Obfuscated Embedding Matrix) E (Embedding Matrix) N_{text} (Optimal Integer Matrix)

Figure 2: Gaussian Periodic Obfuscation in text modality.

4.3 Periodic Activation for Nonlinear Operators

To effectively absorb the periodic perturbations introduced by client-side obfuscation while maintaining stable feature learning, the nonlinear operators in the server-side network need to exhibit sensitivity to periodic patterns. Based on this requirement, we replace conventional nonlinear activations (e.g., ReLU) with periodic activation functions, embedding controllable oscillatory behavior into deep models to interpret and regularize periodic signals originating from obfuscated inputs. However, existing periodic activation functions—such as the pure $\sin(x)$ used in SIREN [34]—while expressive in representing frequency information, often produce dense oscillations that lead to unstable gradients and hinder convergence in deep or large-scale architectures. To address this limitation, we propose *SinLU*, a periodic activation function that integrates sinusoidal periodicity with sparsity by preserving the positive half-wave while suppressing negative oscillations, thereby maintaining periodic expressiveness while significantly enhancing training stability and representational interpretability. Formally, *SinLU* is defined as:

$$\text{SinLU} = \begin{cases} \sin(x) & \text{if } \sin(x) \geq 0 \\ 0 & \text{if } \sin(x) < 0 \end{cases}. \quad (7)$$

Its construction involves two steps: (1) applying $\sin(x)$ to inputs; (2) rectifying negative values to zero. Leveraging the Taylor expansion $\sin(x) = x - \frac{x^3}{6} + \frac{x^5}{120} - \dots$, *SinLU* inherits the

multi-scale representation capacity of sinusoidal bases, where factorial decay $\frac{1}{n!}$ ensures convergence. This enables: single-layer generation of infinite-dimensional feature spaces via polynomial combinations, and stacked-layer approximation of arbitrary periodic functions through cascaded harmonics. For a target function $f(x) = \sum_n a_n \sin(nx) + b_n \cos(nx)$, *SinLU* reorganizes basis coefficients for precise functional fitting.

At the gradient propagation level, the derivative $\cos(x)$ introduces a periodic momentum mechanism. In backpropagation, the parameter update $\theta_{t+1} = \theta_t - \eta \nabla_{\theta} \mathcal{L}$ is modulated as:

$$\nabla_{\theta} \mathcal{L} = \frac{\cos(x) \cdot \partial \mathcal{L}}{\partial \text{SinLU}(x)}, \quad (8)$$

where θ denotes parameters of layers, and $\mathcal{L}$ is loss function. This produces alternating regions of gradient amplification ($|\cos(x)| \approx 1$) and attenuation ($|\cos(x)| \approx 0$). Compared to SIREN, which propagates dense oscillatory gradients at all inputs, *SinLU*'s rectification naturally filters out noisy updates in low-signal regions. This stabilizes training, accelerates convergence when updates align with descent directions, and introduces implicit perturbations near flat zones to escape local minima.

Finally, the sparsity of activations stems from the hard cutoff applied to the negative half-wave in *SinLU*. This periodic gating mechanism ensures that only a fraction of the input spectrum is activated, effectively reducing redundant neuron firing and encouraging compact representations across layers. In contrast to SIREN's fully dense activations, which increase computational redundancy and potential overfitting, *SinLU* yields both efficient optimization and more robust feature learning. Thus, *SinLU* provides a principled way to inject periodic priors into deep models while maintaining stability, sparsity, and strong generalization.

4.4 Periodic Constraint on Linear Operators

While the periodic activation introduced in Section 4.3 endows the nonlinear layers with oscillatory sensitivity, the overall periodicity of the network cannot be guaranteed unless the linear operators are also harmonized with the same periodic structure. Without explicit constraints, the weight matrices may distort or destroy periodic correlations propagated from the activation layers, leading to phase inconsistency and unstable feature representations. To ensure global periodic coherence, we introduce a periodic constraint on linear operators that enforces weight alignment with a discrete periodic lattice. We achieve this by constraining the weight matrix W such that its coefficients correspond to integer multiples of a base period, thereby synchronizing the transformations of linear layers with the oscillatory behaviors induced by the nonlinear activations. This design embeds periodicity across both linear and nonlinear components, resulting in stable propagation and consistent periodic representations throughout the network.

Formally, A basic computational block in a DNN can be represented as $F(x) = \text{SinLU}(f(x))$, where $f(x) = Wx + b$. The proposed constraint guides W toward $\frac{mt}{T}$, where T denotes the minimum positive period of the constructed periodic DNN, $t = 2\pi$ is the period of *SinLU*, and $\mathbf{m} \in \mathbb{Z}^d$ is an integer vector.

THEOREM 1. For $F(x) = \text{SinLU}(f(x))$ where $f(x) = Wx + b$. If the weight W satisfies:

$$W = \frac{\mathbf{m}t}{T},$$

for any integer vector $\mathbf{m}$, $F(x)$ is periodic with period T , i.e., $F(x+T) = F(x)$.

The proof of Theorem 1 is presented in Appendix B. This constraint effectively enables the introduction of periodicity into the entire DNNs. The core principle of this constraint is to redefine the coefficients of linear operators such that they satisfy periodic propagation conditions while preserving the model's expressive capacity. For scalar inputs, deviations from ideal periodic weights are penalized by the regularization term

$$\mathcal{L}_{\text{reg}} = \lambda \left\| W - \frac{\mathbf{m}t}{T} \right\|^2, \quad (9)$$

where each $\mathbf{m}_i$ is determined by projecting W_{ii} onto the periodic lattice:

$$\mathbf{m}_i = \text{round}\left(\frac{W_{ii}T}{t}\right) = \arg \min_{\mathbf{m} \in \mathbb{Z}} \left| W_{ii} - \frac{\mathbf{m}t}{T} \right|. \quad (10)$$

In the high-dimensional case, W is constrained to a diagonal-dominant matrix, with diagonal entries aligned to $\frac{\mathbf{m}_i t}{T}$ and off-diagonal terms regularized to suppress cross-dimensional interference:

$$\mathcal{L}_{\text{reg}} = \lambda_1 \left\| \text{diag}(W) - \frac{\mathbf{m}t}{T} \right\|^2 + \lambda_2 \left\| \text{off-diag}(W) \right\|^2. \quad (11)$$

To ensure stable convergence, we adopt a two-stage optimization strategy, as shown in Appendix C. In the **soft periodic warm-up** stage, the weights evolve freely in $\mathbb{R}$ with a relaxed regularization, which encourages alignment while retaining smooth gradients:

$$\mathcal{L}_{\text{stage1}} = \mathcal{L}_{\text{task}} + \lambda_1 \left\| \text{diag}(W) - \frac{\mathbf{m}t}{T} \right\|^2 + \lambda_2 \left\| \text{off-diag}(W) \right\|^2, \quad (12)$$

where $\mathcal{L}_{\text{task}}$ denotes the primary task loss (e.g., cross-entropy), λ_1 and λ_2 are hyperparameters controlling the strength of regularization. After E_{warm} epochs, the **hard periodic quantization** stage enforces exact integer constraints by projection:

$$W^{(t+1)} = \frac{t}{T} \cdot \text{round}\left(\frac{TW^{(t)}}{t}\right), \quad (13)$$

and the loss is updated as

$$\mathcal{L}_{\text{stage2}} = \mathcal{L}_{\text{task}} + \beta \left\| W - \frac{\mathbf{m}t}{T} \right\|^2, \quad (14)$$

where $\beta \gg \lambda_1$ enforces strict lattice alignment. As shown in Appendix C, the DNN period T that satisfies the constraint is the same as the period t of the periodic activation function. This two-stage scheme ensures (i) smooth convergence in the warm-up phase, (ii) exact periodic alignment via lattice projection, and (iii) computational efficiency with $O(N)$ complexity.

In summary, the periodic constraint on linear operators establishes discrete lattice alignment for weights, guaranteeing global periodicity when combined with *SinLU* activations. The two-stage optimization ensures smooth convergence, precise periodic alignment, and $O(n)$ computational complexity. Together, these mechanisms consistently embed periodicity across the network while preserving expressiveness and stability.

4.5 Periodic-aware Contrastive Learning

While the periodic mechanisms in both the client-side obfuscation and the server-side network architecture establish local consistency, the representations of periodically transformed samples may still diverge across different phase shifts. Such discrepancies can lead to unstable optimization and inconsistent feature learning. To address this, we introduce Periodic-aware Contrastive Learning, which enforces representation invariance under periodic transformations and enhances training stability.

Specifically, given an input X , we generate two obfuscated variants X' and X'' by independently applying *Gaussian Periodic Obfuscation*:

$$X' = X + N' \cdot T, \quad X'' = X + N'' \cdot T, \quad (15)$$

ensuring that each obfuscated sample is distributionally aligned with Gaussian noise $s \sim \mathcal{N}(\mu, \sigma^2)$. For each input pair (X', X'') , their representations are extracted as $f(X')$ and $f(X'')$. Similarity is quantified via cosine similarity:

$$s(f(X'), f(X'')) = \frac{f(X')^\top f(X'')}{\|f(X')\| \|f(X'')\|}. \quad (16)$$

The contrastive objective maximizes agreement between the two *Gaussian Periodic Obfuscation*-derived views:

$$\mathcal{L}_{con} = -\frac{1}{P} \sum_{p=1}^P s(f(X'_p), f(X''_p)), \quad (17)$$

where P is the number of paired samples. By aligning representations of Gaussian-periodic variants, *Periodic-aware Contrastive Learning* stabilizes feature learning and reduces gradient fluctuations during training, which leads to faster convergence and higher final accuracy. This consistency constraint allows the model to focus on invariant task-relevant features rather than overfitting to perturbation noise, thereby improving both optimization efficiency and generalization.

5 Evaluation

5.1 Experiment Design and Baselines

Datasets. For image benchmarks, we use the MNIST [4], LFW [26], AT&T [3], and MedMNIST [39], covering digit classification, face recognition, human identity verification, and medical image classification, respectively. For text benchmarks, we use FiQA for financial sentiment analysis, FinNER for financial named entity recognition, and AG News for topic classification. Overall, these datasets provide both general and domain-specific challenges across visual and textual modalities, laying the foundation for a comprehensive evaluation of privacy-preserving representation learning.

Evaluation Metrics. We evaluate *PeriNet* along three dimensions: *utility*, *efficiency*, and *robustness*. For *utility*, we employ task-specific metrics: classification accuracy for digit classification, medical image classification, face recognition, human identity verification, and topic classification; F1-score for financial sentiment analysis; and entity-level F1-score for financial named entity recognition. For *efficiency*, we measure throughput in terms of samples processed per second, capturing the computational overhead introduced by the obfuscation mechanism. For *robustness*, we conduct both qualitative and quantitative privacy assessments: (i) visual and

semantic reconstruction quality under heuristic and GAN-based inversion attempts, and (ii) privacy leakage quantified by brute-force search complexity and attack success rates.

Baseline Models. We compare *PeriNet* against representative privacy-preserving approaches. For vision tasks, baselines include Disguised-Nets [36], Bit-Planes [12], CODER [38], Block-DP [18], and VIM [21]. For text tasks, we select three representative methods: EW-tune [2], 1-Diffractor [28], EP [7]. All baselines are implemented under comparable configurations to ensure fairness in evaluation.

Implementation Details. Experiments are conducted on an NVIDIA GeForce RTX 3090 GPU with Ubuntu 20.04 and PyTorch 1.9. For vision tasks, we evaluate across five representative neural architectures—AlexNet [20], VGG [33], ResNet [13], DenseNet [16], and ViT [6]—to test generality. For text tasks, we adopt a diverse set of architectures, including BiLSTM [24], Text-CNN [31], Transformer encoders (TransEnc) [5], BERT [40], and RoBERTa [29]. Regarding model initialization, BERT and RoBERTa are instantiated using public checkpoints pre-trained on the English Wikipedia and BookCorpus datasets, whereas all other vision and text backbones are trained from scratch. To adapt these architectures to the *PeriNet* paradigm, we implement specific structural modifications: for CNN-based models, the Gaussian Periodic Obfuscation is applied directly to raw input images, with standard ReLU activations replaced by SinLU; for Transformer-based models (including ViT and language models), the obfuscation layer is inserted immediately after the patch or token embedding stage, and GELU/ReLU functions are substituted with SinLU. Across all architectures, periodic lattice constraints are imposed on linear operators. The adapted models are subsequently aligned via Period-aware Contrastive Learning, followed by supervised training or fine-tuning on the respective downstream datasets to ensure robust performance under the periodic obfuscation regime.

5.2 Overall Performance of *PeriNet* in the Image Modality

Utility. To assess the utility preservation of *PeriNet*, we evaluate its classification accuracy across five visual datasets and multiple network architectures. As summarized in Table 2, *PeriNet* consistently achieves near-baseline performance while outperforming existing obfuscation approaches by a clear margin. Specifically, it surpasses Disguised-Nets and Bit-Planes by an average of 3–8%, and maintains accuracy within 1–2% of the non-obfuscated upper bound on most datasets. This indicates that the proposed periodic obfuscation preserves task-relevant semantics and stable representational structure, even under strong privacy constraints.

Throughput. Table 3 reports the throughput (images per second) across architectures. *PeriNet* exhibits comparable or superior performance to prior methods, achieving up to 7501 images per second on AlexNet and maintaining over 95% of the non-obfuscated throughput on all tested models. The slight overhead primarily stems from the sinusoidal activation and periodic constraints, yet remains far lower than the cost of traditional noise-based or block-transform obfuscations. These results demonstrate that *PeriNet* achieves a balanced trade-off between privacy, accuracy, and efficiency—providing strong protection without compromising practicality in large-scale deployments.

Table 2: Comparison of classification accuracy across datasets using different obfuscation methods.

Method	LFW ↑					MNIST ↑					AT&T ↑					CIFAR-10 ↑					MedMNIST ↑				
	AlexNet	VGG	ResNet	DenseNet	ViT	AlexNet	VGG	ResNet	DenseNet	ViT	AlexNet	VGG	ResNet	DenseNet	ViT	AlexNet	VGG	ResNet	DenseNet	ViT	AlexNet	VGG	ResNet	DenseNet	ViT
Baseline (No Defense)	70.23%	90.31%	94.25%	84.21%	94.59%	86.28%	99.51%	99.27%	98.73%	99.87%	80.24%	99.23%	99.73%	99.01%	99.87%	78.15%	92.64%	96.25%	88.35%	98.17%	84.12%	92.31%	94.85%	91.27%	96.02%
Disguised-Nets [36]	66.18%	85.27%	86.37%	78.68%	90.26%	84.44%	97.51%	97.23%	96.26%	97.57%	78.91%	98.51%	98.13%	97.86%	98.89%	74.18%	89.34%	90.28%	84.56%	92.29%	80.26%	88.47%	90.65%	87.34%	92.13%
Bit-Planes [12]	60.26%	83.31%	81.64%	76.57%	86.98%	80.18%	95.63%	94.14%	94.58%	96.23%	76.43%	95.63%	96.57%	94.24%	95.34%	68.98%	87.63%	84.32%	81.45%	88.75%	77.54%	86.25%	88.34%	85.46%	90.08%
CODER [38]	52.22%	70.23%	70.53%	60.64%	72.17%	67.78%	69.34%	61.36%	83.26%	79.51%	62.46%	80.34%	78.36%	80.48%	80.61%	60.12%	80.71%	81.12%	78.33%	86.21%	69.83%	74.92%	76.37%	72.64%	78.15%
Block-DP [18]	57.74%	76.48%	76.36%	68.82%	78.61%	74.92%	84.61%	83.84%	83.17%	85.47%	69.11%	84.28%	82.64%	84.57%	84.67%	67.88%	79.68%	81.37%	73.65%	84.97%	73.12%	79.34%	80.61%	77.45%	82.02%
VIM [21]	60.51%	81.28%	83.51%	69.01%	81.87%	85.92%	98.12%	98.81%	98.03%	99.14%	71.21%	85.74%	83.57%	82.33%	83.23%	74.51%	90.28%	92.11%	85.23%	93.98%	79.01%	87.92%	89.74%	86.23%	91.11%
PeriNet	68.32%	88.36%	92.73%	81.58%	91.57%	86.05%	99.34%	99.02%	98.72%	99.08%	79.54%	99.01%	99.14%	98.86%	99.63%	75.32%	91.32%	93.73%	86.82%	96.54%	83.26%	91.87%	93.42%	90.12%	95.01%

Table 3: Comparison of throughput (images per second) across different models using various obfuscation methods.

Method	AlexNet	VGG	ResNet	DenseNet	ViT
Baseline (No Defense)	7636.8	421.4	632.2	565.2	325.2
Disguised-Nets [36]	7223.2	405.3	595.7	527.1	307.2
Bit-Planes [12]	7133.8	397.2	586.5	519.8	299.6
CODER [38]	7338.7	410.7	612.1	543.1	304.1
Block-DP [18]	6841.5	403.2	596.7	519.2	314.4
VIM [21]	6810.1	401.9	582.2	520.4	316.5
PeriNet	7501.1	419.1	621.8	556.5	320.8

5.3 Performance of PeriNet in the Text Modality

Utility. To evaluate the effectiveness of *PeriNet* in textual tasks, we conduct experiments on three representative datasets using diverse model backbones. As shown in Table 4, *PeriNet* consistently achieves near-baseline performance across all architectures, outperforming existing privacy-preserving methods such as EW-tune, 1-Diffractor, and EP. Specifically, the relative performance drop compared to the non-obfuscated baseline remains within 1–2%, while other approaches exhibit losses of 5–10% or more. This demonstrates that the proposed periodic obfuscation effectively conceals sensitive textual content while maintaining semantic integrity and downstream task performance.

Throughput. Table 7 further reports the throughput (sentences per second) under various models. *PeriNet* maintains over 90–95% of the original throughput across all architectures, significantly outperforming EW-tune, which incurs a substantial computational cost due to gradient clipping and noise sampling. Even on large-scale transformer models such as BERT and RoBERTa, the proposed method introduces only marginal overhead (less than 5%), confirming its scalability and efficiency.

5.4 Security Evaluation on Adversarial Attacks

Brute-force Attack Validation. We analyze the combinatorial hardness of exhaustive-search attacks in a modality-agnostic way. Let N denote the total number of independent elements in a sample — for example, $N = w \cdot h$ for a $w \times h$ image, or $N = L \cdot d$ for a text embedding matrix of length L and embedding dimension d . Each element is perturbed by an integer multiplier $k \in [-K, K] \cap \mathbb{Z}$, giving $C = 2K + 1$ possible states per element. The adversary’s naive search space is:

$$S = C^N, \quad (18)$$

which grows exponentially with N .

From an information-theoretic viewpoint, the brute-force search space corresponds to a Shannon entropy of:

$$H_{\text{brute}} = N \cdot \log_2 C \quad (\text{bits}). \quad (19)$$

As concrete illustrations, with $K = 4$ ($C = 9$) a 64×64 image has $N = 4096$ and thus:

$$H_{\text{brute}} \approx 4096 \cdot \log_2 9 \approx 13,049 \text{ bits}, \quad (20)$$

while a token embedding matrix with $L = 128$ and $d = 768$ has $N = 128 \cdot 768 = 98,304$ and

$$H_{\text{brute}} \approx 98,304 \cdot \log_2 9 \approx 311,616 \text{ bits}. \quad (21)$$

Covering just 1% of the solution space requires about 10^{3872} guesses, which is computationally, energetically, and temporally infeasible. Under Gaussian periodic obfuscation, the brute-force space grows exponentially with matrix size, and even with full knowledge of matrix dimensions and period, reconstruction remains intractable. This confirms strong defense from both combinatorial and information-theoretic perspectives.

Validation of Model Inversion Attacks in Image Modality.

To further evaluate the ability of *PeriNet* to defend against model inversion attacks aimed at recovering original images, a GAN-based attack experiment was designed [14]. GAN attacks typically attempt to deceive the model using false labels in order to steal sensitive information or disrupt normal operations. In our experiment, a GAN is employed to generate adversarial samples from obfuscated images, attempting to reconstruct the original data using fake labels. As shown in Figure 7, the experimental results demonstrate that the effectiveness of GAN attacks is significantly reduced after applying our obfuscation method, and the attacker is unable to successfully reconstruct the original feature maps from the generated adversarial samples. This indicates that the proposed obfuscation approach provides effective defense against advanced GAN-based attacks, further enhancing the overall system security.

Validation of Model Inversion Attacks in Text Modality.

To assess the robustness of *PeriNet* against textual inversion threats, we evaluate two representative attack types aimed at reconstructing raw text from obfuscated embeddings. The first is a nearest-neighbor decoding attack, where the adversary decodes embeddings over a surrogate vocabulary to retrieve the most probable tokens. The second is an inverse mapping model attack, where a light-weight neural decoder is trained to recover text sequences from obfuscated representations. Reconstruction quality is measured by BLEU (surface-level n-gram overlap), BERTScore (semantic similarity in embedding space), and PII recovery rate (sensitive-entity leakage). As shown in Table 5 and Table 6, *PeriNet* consistently achieves the lowest BLEU, BERTScore, and PII recovery across both attacks—reducing semantic and entity-level recoverability by more than 70% compared to EP and CODER baselines. The reconstructed sentences predominantly reflect the structured noise patterns introduced by Gaussian periodic obfuscation rather than the original content, confirming that *PeriNet* effectively suppresses both lexical and semantic leakage in textual representations.

Table 4: Comparison of task performance with different obfuscation methods across text datasets.

Method	FiQA (F1-score) $\uparrow$					FinNER (F1-score) $\uparrow$					AG News (Accuracy) $\uparrow$				
	BiLSTM	Text-CNN	TransEnc	BERT	RoBERTa	BiLSTM	Text-CNN	TransEnc	BERT	RoBERTa	BiLSTM	Text-CNN	TransEnc	BERT	RoBERTa
Baseline (No Defense)	75.5	76.5	78.0	78.2	80.6	79.2	81.3	82.5	83.4	84.6	90.8	92.1	93.5	94.8	95.2
EW-tune ($\epsilon = 1$) [2]	66.3	69.5	70.8	70.3	71.2	70.1	71.4	72.8	74.2	75.3	83.2	84.7	86.4	88.2	89.0
1-Diffractor (15%) [28]	69.5	71.1	72.6	72.5	73.8	69.3	70.8	71.4	71.8	72.2	84.0	85.2	87.5	89.5	90.1
EP ($\sigma = 0.1$) [7]	69.2	74.0	75.1	74.9	75.3	73.2	74.1	75.6	77.9	78.5	86.1	87.3	88.8	91.3	91.7
PeriNet	74.2	75.8	77.1	77.6	79.4	78.2	79.7	80.9	82.1	83.0	89.7	91.0	92.2	93.7	94.1

Table 5: Model inversion attack results under Nearest-Neighbor Decoding Attack in text modality.

Method	Nearest-Neighbor Decoding Attack		
	BLEU $\downarrow$	BERTScore $\downarrow$	PII Recovery (%) $\downarrow$
Baseline (No Defense)	48.7	0.82	69.5
EW-tune ($\epsilon = 1$) [2]	36.8	0.72	40.3
1-Diffractor (15%) [28]	33.4	0.69	35.1
EP ($\sigma = 0.1$) [7]	28.1	0.63	27.5
PeriNet	20.3	0.45	11.2

Table 6: Model inversion attack results under Inverse Mapping Model Attack in text modality.

Method	Inverse Mapping Model Attack		
	BLEU $\downarrow$	BERTScore $\downarrow$	PII Recovery (%) $\downarrow$
Baseline (No Defense)	52.8	0.87	74.2
EW-tune ($\epsilon = 1$) [2]	38.6	0.74	41.5
1-Diffractor (15%) [28]	34.2	0.71	36.8
EP ($\sigma = 0.1$) [7]	29.5	0.65	28.7
PeriNet	17.9	0.42	9.6

Table 7: Throughput (sentences per second) of different text models using various obfuscation methods.

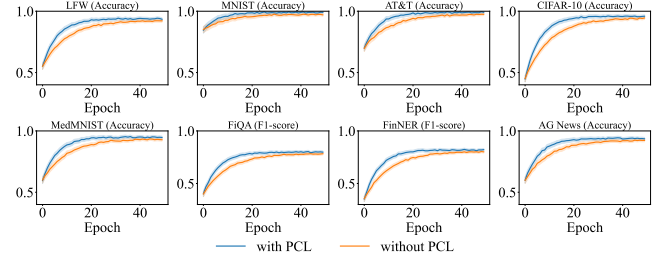
Method	BiLSTM	Text-CNN	TransEnc	BERT	RoBERTa
Baseline (No Defense)	4821.3	5210.6	4387.2	1250.4	1189.5
EW-tune ($\epsilon = 1$) [2]	3112.8	3520.5	2947.3	720.2	683.7
1-Diffractor (15%) [28]	3921.4	4102.8	3679.1	980.3	935.6
EP ($\sigma = 0.1$) [7]	4052.7	4289.4	3802.5	1020.7	974.1
PeriNet	4685.6	4982.1	4233.9	1200.8	1148.2

5.5 Ablation on Periodic-aware Contrastive Learning

We conduct an ablation experiment to evaluate the effect of periodic-aware contrastive learning. Models trained with and without Periodic-aware Contrastive Learning share the same backbone and obfuscation settings. As shown in Figure 3, the results show that incorporating periodic-aware contrastive learning notably accelerates convergence and improves final accuracy, demonstrating that enforcing invariance to periodic augmentations enhances representation stability and generalization in privacy-preserving training.

6 Discussion

The proposed *PeriNet* is readily extensible beyond vision and text to other modalities such as speech, sensor, or biomedical signals. For these data types, their embeddings can be directly processed

**Figure 3: Ablation on periodic-aware contrastive learning (PCL). The accuracy and F1-score curves demonstrate that incorporating the proposed contrastive objective accelerates convergence and improves the final performance.**

through Gaussian periodic obfuscation, while corresponding networks adopt periodic architectures by replacing nonlinear activations with *SinLU* and applying periodic constraints to linear operators. With periodicity-aware contrastive learning for efficient optimization, such models can achieve secure and low-loss training and inference across heterogeneous modalities. This generality highlights the potential of periodic obfuscation as a unified and practical paradigm for privacy-preserving deep learning.

7 Conclusion

We propose *PeriNet*, a modality-agnostic privacy-preserving framework that integrates periodicity into both data obfuscation and DNNs design. On the client side, it introduces a *Gaussian Periodic Obfuscation* mechanism that applies structured periodic perturbations to protect sensitive content, while on the server side, it constructs a periodic DNN framework with sinusoidal activations and periodic constraints to ensure stable and consistent representations. Extensive experiments on vision and text benchmarks show that *PeriNet* resists reconstruction and inversion attacks while maintaining near-baseline performance, achieving 99.34% accuracy on vision tasks and 94.1% on language tasks, outperforming the strongest privacy baselines by up to 8% with minimal computational overhead.

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Appendix

A Notation

We list the notations in Table 8.

B Proof of Theorem 1

Let the linear transformation be defined as $f(x) = Wx + b$, where W satisfies the periodic weight constraint

$$WT = m\mathbf{t}, \quad \mathbf{m} \in \mathbb{Z}.$$

Under this condition, for any input shift by T , we have

$$\begin{aligned} f(x + T) &= W(x + T) + b \\ &= Wx + WT + b \\ &= Wx + b + m\mathbf{t} \\ &= f(x) + m\mathbf{t}. \end{aligned}$$

Applying the periodic activation function $\text{SinLU}(\cdot)$ with period t , we obtain

$$\begin{aligned}\text{SinLU}(f(x + T)) &= \text{SinLU}(f(x) + mt) \\ &= \text{SinLU}(f(x)).\end{aligned}$$

Therefore, each linear-activation block satisfies

$$\text{SinLU}(f(x + T)) = \text{SinLU}(f(x)),$$

which establishes the periodic invariance property claimed in Theorem 1. $\square$

C Proof of Global Periodicity and Backpropagation Invariance

Theorem 2 (Global Periodicity of PeriNet). Let the DNN $F(x)$ be composed of L layers, each defined as

$$F(x) = f_L \circ f_{L-1} \circ \dots \circ f_1(x),$$

where each $f_i(x) = \sigma_i(W_i x + b_i)$, σ_i is the periodic activation SinLU with fundamental period $t = 2\pi$, and W_i satisfies the periodic lattice constraint $W_i T = m_i t$ for some integer matrix m_i . Then, for any integer k ,

$$F(x + kT) = F(x).$$

Proof. For the first layer,

$$f_1(x + T) = \sigma_1(W_1(x + T) + b_1) = \sigma_1(W_1 x + b_1 + W_1 T).$$

By the weight constraint $W_1 T = m_1 t$, and the periodicity $\sigma_1(z + t) = \sigma_1(z)$,

$$f_1(x + T) = \sigma_1(W_1 x + b_1 + m_1 t) = \sigma_1(W_1 x + b_1) = f_1(x).$$

Assume $f_j(x + T) = f_j(x)$ holds for the j -th layer. Then for layer $j + 1$,

$$f_{j+1}(x + T) = \sigma_{j+1}(W_{j+1} f_j(x + T) + b_{j+1}) = \sigma_{j+1}(W_{j+1} f_j(x) + b_{j+1}) = f_{j+1}(x),$$

so by induction $F(x + T) = F(x)$. Thus, the network F is globally periodic with period T under the alignment $t = T$. $\square$

Gradient Consistency and Backpropagation Invariance.

Let the total loss be $L = \mathcal{L}(F(x), y)$. During backpropagation, the gradient with respect to W_i is

$$\frac{\partial L}{\partial W_i} = \frac{\partial L}{\partial F(x)} \frac{\partial F(x)}{\partial W_i} = \frac{\partial L}{\partial F(x + T)} \frac{\partial F(x + T)}{\partial W_i}.$$

From the theorem, $F(x + T) = F(x)$; hence $\frac{\partial L}{\partial F(x + T)} = \frac{\partial L}{\partial F(x)}$. Next, the Jacobian term satisfies

$$\frac{\partial F(x + T)}{\partial W_i} = \frac{\partial}{\partial W_i} \sigma_i(W_i(x + T) + b_i) = \sigma'_i(W_i(x + T) + b_i) (x + T)^\top.$$

Since $\sigma'_i(z) = \cos(z)$ is itself periodic with the same period $t = T$, we have $\sigma'_i(W_i(x + T) + b_i) = \sigma'_i(W_i x + b_i)$. Therefore,

$$\frac{\partial F(x + T)}{\partial W_i} = \sigma'_i(W_i x + b_i) x^\top = \frac{\partial F(x)}{\partial W_i},$$

and consequently,

$$\boxed{\frac{\partial L}{\partial W_i}(x + T) = \frac{\partial L}{\partial W_i}(x)}.$$

This equality implies that all gradients, updates, and backpropagation steps are invariant under period T . The training dynamics remain identical for x and $x + T$, ensuring that the introduction of

periodic constraints and SinLU activations does not disturb optimization.

Corollary (Compatibility with Contrastive Learning). Under the same periodic invariance, the periodic contrastive loss L_{con} defined in Eq.(17) satisfies $L_{\text{con}}(x + T) = L_{\text{con}}(x)$, since both positive and negative representations $f(x + T)$ and $f(x)$ are equivalent. Thus, the global gradient field for both the task and contrastive objectives remains unchanged, preserving convergence properties.

Conclusion. The above results demonstrate that the periodic constraint $W_i T = m_i t$ and the SinLU activation together ensure global periodic invariance in both the forward and backward passes. Consequently, PeriNet achieves phase-aligned stability without affecting optimization or gradient propagation, thereby maintaining convergence while providing structural periodicity.

D Gaussian Periodic Obfuscation Explanation and Effect Demonstration on Image Data

As illustrated in Figure 4, the processing pipeline of the Gaussian periodic obfuscation mechanism on image data is presented, while Figure 5 further visualizes the obfuscation effects after applying our method and others.

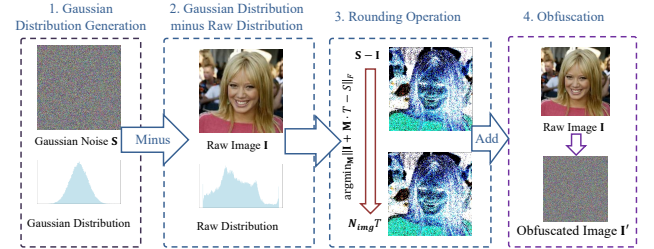


Figure 4: Gaussian Periodic Obfuscation in image modality.

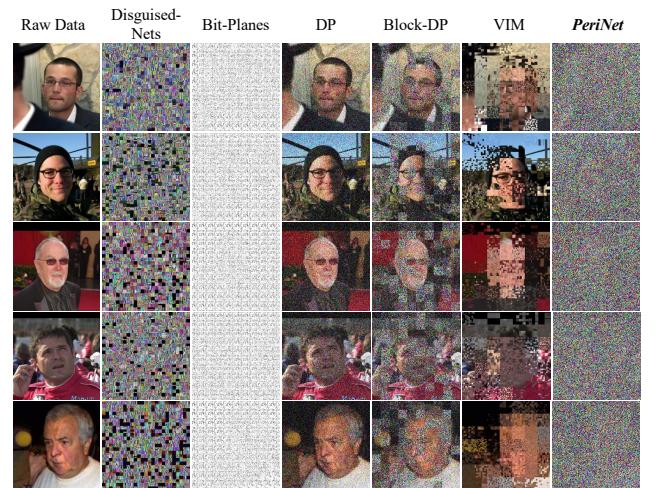


Figure 5: Visualization of obfuscated images by different obfuscation methods.

Algorithm 1: PeriNet training strategy

Input: Original data matrix X (e.g., image pixels or text embeddings), period T , hyperparameters $\lambda_1, \lambda_2, \beta$, E_{warm}

Output: Trained periodic DNN model F

// Client-side: Gaussian Periodic Obfuscation

- 1 **for** each element $x_{i,j}$ in X **do**
- 2 Sample target value $s_{i,j} \sim \mathcal{N}(\mu, \sigma^2)$;
- 3 Compute optimal integer $q_{i,j} = \arg \min_q |x_{i,j} + q \cdot T - s_{i,j}|$;
- 4 Generate obfuscated element $x'_{i,j} = x_{i,j} + q_{i,j} \cdot T$;
- 5 **end**
- 6 Obtain obfuscated data X' ;
- // Server-side: Periodic DNN Training
- 7 Initialize network F with *SinLU* activation and periodic constraints;
- 8 **for** $e = 1$ to E_{max} **do**
- 9 Forward pass: $\hat{y} = F(X')$;
- 10 Compute task loss $\mathcal{L}_{task}$;
- // Periodic constraint loss (two-stage training)
- 11 **if** $e < E_{warm}$ **then**
- 12 // Stage 1: Soft periodic warm-up
- 13 Compute integer projection vector:
 $\mathbf{m} = \text{round}\left(\frac{T \cdot \text{diag}(W)}{t}\right)$;
- 14 Compute soft constraint loss:
 $\mathcal{L}_{reg} = \lambda_1 \|\text{diag}(W) - \frac{t \cdot \mathbf{m}}{T}\|^2 + \lambda_2 \|\text{offdiag}(W)\|^2$;
- 15 **else**
- 16 // Stage 2: Hard periodic quantization
- 17 Update weights: $W \leftarrow \frac{t}{T} \cdot \text{round}\left(\frac{T \cdot W}{t}\right)$;
- 18 Compute hard constraint loss:
 $\mathcal{L}_{reg} = \beta \cdot \|W - \frac{t}{T} \cdot \text{round}\left(\frac{T \cdot W}{t}\right)\|^2$;
- 19 **end**
- // Period-aware contrastive learning
- 20 Generate periodic augmentation $X'' = X + n' \cdot T$ (with random integer q');
- 21 Extract features $f(X')$ and $f(X'')$;
- 22 Compute contrastive loss $\mathcal{L}_{con} = -\cos(f(X'), f(X''))$;
- // Total loss
- 23 $\mathcal{L}_{total} = \mathcal{L}_{task} + \mathcal{L}_{reg} + \mathcal{L}_{con}$;
- 24 Backpropagate and update F 's parameters;
- 25 **end**

E Case Study

This section analyzes the periodic DNNs construction method proposed in Section 4, focusing on the implementation of the proposed periodic DNNs construction method on two representative and structurally distinct architecture, as shown in Figure 6.

Enhancing Periodicity in CNNs. For CNN architectures, *PeriNet* integrates periodic obfuscation compatibility via key modifications, as shown in Figure 6 (a): first, replacing ReLU activations

with *SinLU* for periodic nonlinearity. Second, periodic constraints on convolutional kernels $W \in \mathbb{R}^{k \times k \times c_{in} \times c_{out}}$ enforcing diagonal dominance:

$$Loss_{reg}^{conv} = \sum_{i=1}^{C_{out}} \sum_{j=1}^{C_{in}} \lambda \|W^{(i,j)} - \text{diag}(W^{(i,j)})\|_F^2 + \sum_{i=1}^{C_{out}} \sum_{j=1}^{C_{in}} \lambda \left\| \text{diag}(W^{(i,j)}) - \frac{\mathbf{m}t}{T} \right\|_2^2.$$

Here, $\mathbf{m}$ is the learned integer multiple of the period, and $F(x)$ is a residual block. Convolutional weights are guided to satisfy periodic constraints, while residual connections are adjusted for phase consistency to preserve periodic behavior. The implementation for $F(x)$ is as follows:

$$x_{l+1} = \text{SinLU}(F(x_l) + x_l), \quad (22)$$

where x_l represents the input of the l -th layer. Experimental results show that CNN with *PeriNet* achieves 93.73% visual privacy protection accuracy on CIFAR-10, with only a minimal 2.52% drop in classification accuracy compared to the baseline. This demonstrates the framework's ability to balance privacy and utility effectively.

Enhancing Periodicity in Transformer. Enhancing periodicity in Transformers involves modifying attention modules, as illustrated in Figure 6 (b). As in CNNs, ReLU activations in feed-forward layers are replaced with *SinLU*, and periodic constraints are applied to patch embedding convolutions. To preserve periodicity in attention scores, the self-attention query and key projections (W_Q and W_K) are constrained as follows:

$$Loss_{reg}^{attention} = \|\text{vec}(W_Q W_K^T) - \frac{\mathbf{m}t}{T}\|^2, \quad (23)$$

This ensures that the attention scores, computed via $\text{softmax}\left(\frac{QK^T}{\sqrt{d}}\right)$, retain periodicity through cosine similarity. Constraining Q and K preserves input periodicity, enabling the PeriNet-equipped Transformer to achieve 96.54% accuracy with only a 1.63% drop in performance.

F Supplementary Experimental Figures

Figure 7 provides visual results of reconstruction attacks.

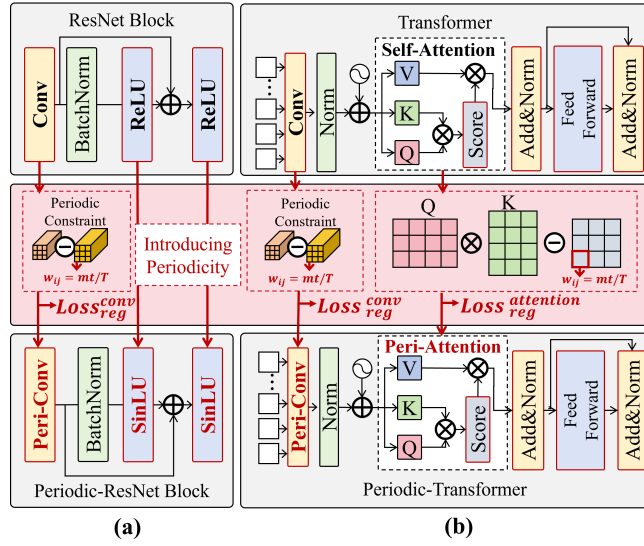


Figure 6: Introducing periodicity in (a) CNN-based model and (b) Transformer-based model.

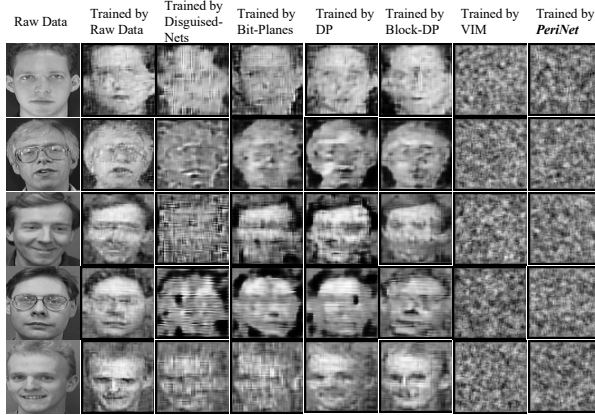


Figure 7: Effect of GAN-based data reconstruction attack with different obfuscation methods.

Table 8: Notation Summary

Symbol	Definition
X	Input tensor
X'	Obfuscated tensor
X''	Obfuscated tensor variations
x	Elements of input tensor
N	Optimal integer matrix
n	Element of integer matrix N
S	Noise tensor
s	Element of noise tensor
$\mathcal{S}$	Solution space size
N	Total number of independent elements in a sample
μ	Distribution mean
σ	Standard deviation
M	An arbitrary integer matrix
m	An integer in M
I	Original image
I'	Obfuscated image
N_{img}	Integer matrix matching the input image
$i_{x,y}$	Original pixel value
$i'_{x,y}$	Obfuscated pixel value
w	Image width
h	Image height
E	Text embedding matrix
E'	Obfuscated embeddings
N_{text}	Integer matrix matching the input text embedding
L	Sequence length
d	Embedding dimension
T	Minimum period of DNN
$F(\cdot)$	DNN block function
$f(\cdot)$	Linear operator
W	Weight matrix
w_{ii}	Elements of weight matrix
b	Bias term
t	Minimum period of SinLU
$\mathcal{L}$	Loss function
θ	Parameters of layers in backpropagation
$\mathbf{m}$	Weight constraint integer vector
$\mathbf{m}_i$	Elements of weight constraint integer vector
$\mathcal{L}_{\text{reg}}$	Regularization loss function
$\mathcal{L}_{\text{task}}$	Primary task loss function
$\mathcal{L}_{\text{con}}$	Loss function in contrast
η	Learning rate
λ	Regularization coefficient
β	Hard constraint coefficient
P	Number of contrastive learning input pairs
p	Index of contrastive learning input pairs
$\sigma(\cdot)$	Periodic activation function
K	Key matrix in self-attention
Q	Query matrix in self-attention
H_{brute}	Minimum entropy
E_{warm}	Warm-up epochs
E_{max}	Total epochs